LINHAO BAI

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PROFESSIONAL EXPERIENCE

Tesla - Robotics Integrations Engineer Sep 2026 - Present

Austin, TX | Mobile Robotics / Overmind

- Work on large-scale AMR, AGV, and ASRS systems covering multi-robot coordination, task sequencing, fleet orchestration, production workflows, and robot-to-equipment interfaces.
- Investigate behavior across distributed services, Kubernetes, NATS messaging, observability, robot runtimes, localization, and industrial PLC / OPC UA systems.
- Work with simulation and virtual commissioning to test task logic, operational scenarios, and system behavior before and during production deployment.
- Develop the design of AI-agent-assisted diagnostics that connect robot incidents, logs, system state, configurations, and source-level evidence.

Georgia Tech STAR Lab - Graduate Research Assistant Oct 2024 - May 2026

Atlanta, GA

- Conducted robot-learning research in dexterous manipulation, visuomotor behavior models, residual reinforcement learning, and sim-to-real transfer.
- Built ROS-based real-time learned-policy execution for a Kinova arm and dexterous hands, including synchronized commands, joint feedback, and experiment logging.

SEER Robotics - Software Engineer / Project Manager Jun 2022 - Jul 2024

Shanghai, China

- Developed C++ robot-controller tools for telemetry, time synchronization, monitoring, OTA / backup, and robot software interfaces.
- Worked on international AMR fleets and a five-AMR / 100+ rack warehouse system: routing, rack reservations, queueing, SLAM / QR correction, and randomized fleet tests.

SELECTED RESEARCH & PUBLICATIONS

Going with the Flow (K-UBM) | CoRL 2026 | Equal co-second author [paper]
Co-authored Koopman-based visuomotor behavior models and event-triggered replanning for dexterous manipulation.

Video2Sim2Real | 2026 preprint [paper]
Co-authored a video-to-simulation-to-real pipeline for autonomous dexterous skill acquisition, involving perception, simulation, and policy learning.

A 1/R Law for Kurtosis Contrast in Balanced Mixtures | IEEE SPL 2026
A signal processing publication on kurtosis contrast in balanced mixtures.

SELECTED PROJECTS

Multi-Robot Pathfinding & Scheduling Simulation
Reproduced published multi-agent planning methods in simulation; studied route conflicts, collision avoidance, and fleet coordination.

SO-101 & MicroDuck | Robot Learning Reproductions
Reproduced open-source manipulation and locomotion learning experiments involving policy training, reinforcement learning, and simulation.

EDUCATION

Georgia Institute of Technology - M.S. Computer Science 2024 - 2026
GPA: 4.0 / 4.0

Chongqing Jiaotong University - B.Eng. Water Resources Engineering 2018 - 2022

TECHNICAL SKILLS

Python, C++, PyTorch, ROS / ROS2, MuJoCo, Linux, Docker, Kubernetes, Git, NATS, OPC UA, SLAM, multi-agent planning, imitation learning, reinforcement learning, Koopman dynamics, sim-to-real, distributed systems.